

Explainable Machine Learning Models for Early Warning of Cholera Outbreaks in Kenya Using Climate and Epidemiological Indicators

By

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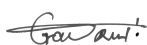
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DECLARATION

This work was carried out at AIMS Rwanda in partial fulfilment of the requirements for a Master of Science Degree.

I hereby declare that, except where due acknowledgement is made, this work has never been presented wholly or in part for the award of a degree at AIMS Rwanda or any other University.

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DEDICATION

I dedicate this work to my family, whose encouragement, patience and unwavering support have sustained me throughout my academic journey.

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I am deeply grateful to my supervisor, Dr. Amos Kipkorir Langat, for his guidance, constructive criticism and continued support throughout the development of this research. His comments and scientific insight were invaluable in strengthening both the analytical approach and the final presentation of this work.

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I extend my appreciation to all those involved in the collection, preparation and documentation of the data used in this study. I am particularly grateful for the surveillance and public-health information that made the analysis of cholera early warning across Kenya possible.

Finally, I am grateful to my family and friends for their encouragement, understanding and support throughout this period of study.

Abstract

Kenya continues to experience recurrent cholera outbreaks, with substantial variation in burden across counties and over time. This study developed and evaluated a county-level early-warning framework for predicting cholera outbreaks two to four weeks in advance across all 47 Kenyan counties. Weekly data from 2015–2025 were integrated across epidemiological, climatic, environmental, demographic and water, sanitation and hygiene (WASH) domains. For the primary analysis, an outbreak was defined as a future county-week with at least five reported cholera cases. A broader definition that additionally classified county-weeks with incidence of at least one case per 100,000 population as outbreaks was examined in sensitivity analysis. The main three-week analysis included 26,508 complete county-week observations.

Separate ARIMA models fitted to each county's weekly cholera case series were used as the statistical baseline, while Random Forest and XGBoost were evaluated as machine-learning classifiers. Model development followed a chronological design: 2015–2022 data were used for training, 2023–2024 for validation, and 2025 was reserved for independent testing. All analyses were conducted in Python. The final XGBoost model achieved an accuracy of 0.7735, sensitivity of 0.5858, specificity of 0.8517, balanced accuracy of 0.7187, ROC AUC of 0.7601 and PR AUC of 0.5933 on the 2025 test period.

Performance at two and three weeks was similar, whereas discrimination declined at four weeks. The three-week horizon was therefore retained because it provided an additional week of lead time with little loss in predictive performance compared with the two-week horizon. When evaluated on the same 2025 observations and using the same outbreak definition, ARIMA produced fewer false alerts and higher specificity, while XGBoost detected more outbreaks. The two approaches therefore showed different operational error profiles rather than one model consistently outperforming the other. SHAP analysis was used as a secondary interpretive tool and indicated that temporal information, recent cholera activity and structural vulnerability contributed strongly to XGBoost predictions.

Overall, the findings demonstrate moderate predictive ability for county-level cholera outbreak risk three weeks in advance. Machine-learning models may therefore provide a useful supplementary early-warning signal for cholera preparedness in Kenya, but they should complement rather than replace routine surveillance, laboratory confirmation and epidemiological judgement. Prospective validation and periodic updating would be required before operational deployment.

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1. Introduction

This chapter introduces the study, presents the research problem, aim, objectives and research questions, and outlines the significance and scope of the study.

1.1 Background of the Study

Cholera is an acute diarrhoeal disease caused by toxigenic *Vibrio cholerae*, commonly transmitted through food or water contaminated with faecal material (World Health Organization, 2024; Ali et al., 2015). Severe infection can lead to rapid dehydration and death without timely treatment (World Health Organization, 2024). Although cholera is preventable through safe water, sanitation, hygiene, vaccination and effective outbreak response, it remains an important public-health problem in many endemic settings (World Health Organization, 2024; Global Task Force on Cholera Control, 2017). An estimated 1.3–4.0 million cases and 21,000–143,000 deaths occur annually in endemic countries, with Sub-Saharan Africa carrying a substantial proportion of the burden (Ali et al., 2015; Lessler et al., 2018).

Cholera transmission is closely associated with inadequate water and sanitation, while crowding and population movement may increase opportunities for exposure (Cowman et al., 2017; Lessler et al., 2018; Ngere et al., 2024). Climatic and environmental conditions may also influence cholera transmission (Cann et al., 2013; Jutla et al., 2013; Tompkins and Di Giuseppe, 2015). Heavy rainfall and flooding may overwhelm drainage and sanitation systems and contribute to faecal contamination of water sources (Cann et al., 2013; Jutla et al., 2013; Reyburn et al., 2011). Where contaminated water is subsequently used for drinking, cooking or other household purposes, the risk of exposure to *V. cholerae* may increase (Cann et al., 2013; Jutla et al., 2013; Tompkins and Di Giuseppe, 2015). Temperature has also been associated with environmental conditions favourable to the persistence of *V. cholerae*, although climatic relationships vary across settings and over time (Reyburn et al., 2011; Perez-Saez et al., 2017).

Figure 1.1 provides a conceptual illustration of one possible pathway through which heavy rainfall, inadequate sanitation and contaminated water may contribute to cholera transmission.

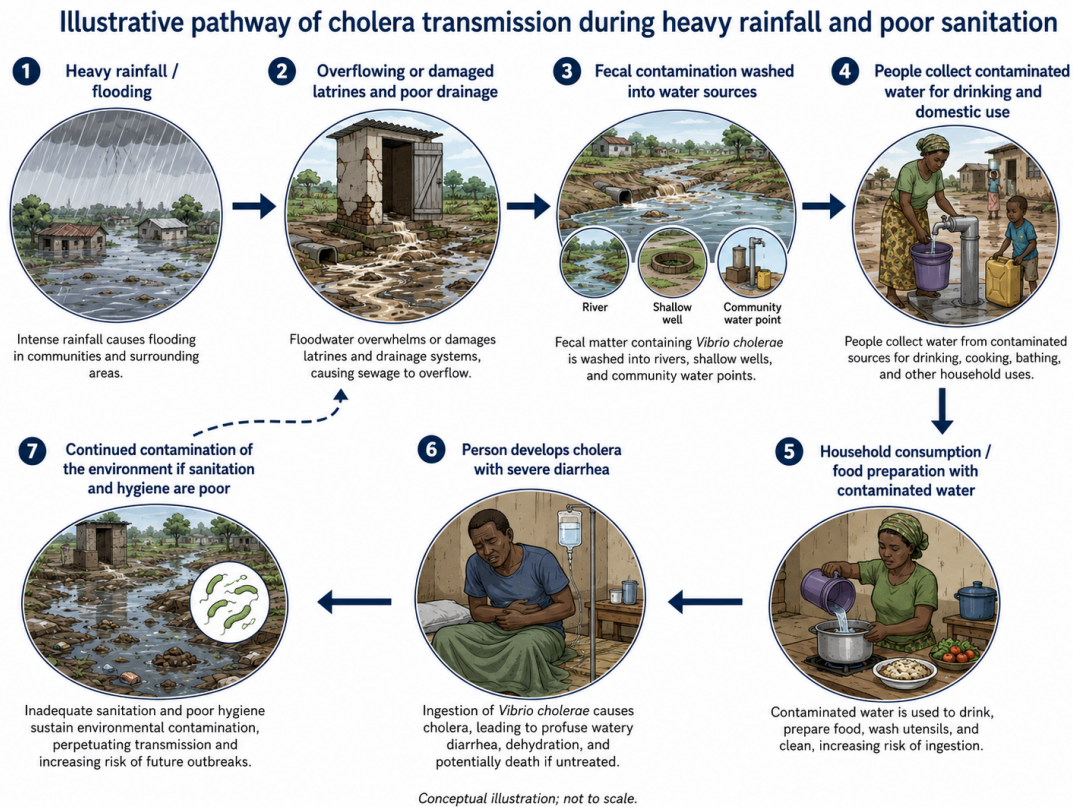


Figure 1.1: Conceptual illustration of a possible cholera transmission pathway during heavy rainfall, flooding and inadequate sanitation. The figure is illustrative and does not represent an empirically observed transmission sequence. Developed for this study and informed by published evidence on waterborne cholera transmission and environmental risk (World Health Organization, 2024; Cann et al., 2013; Jutla et al., 2013).

Kenya has experienced recurrent cholera outbreaks with substantial geographical and temporal variation (Mutonga et al., 2013; George et al., 2016; Kiama et al., 2023). Historical surveillance has documented outbreaks in urban, coastal, lake, arid and refugee settings (Mutonga et al., 2013). More recent studies have also identified geographical differences in cholera burden and WASH vulnerability (Kiama et al., 2023; Boru et al., 2023), while outbreak investigations have highlighted the influence of crowding, population movement and limitations in outbreak response (Curran et al., 2018; Ngere et al., 2024).

The study dataset similarly demonstrates substantial spatial variation in reported cholera burden across the country. Figure 1.2 shows cumulative reported cholera cases across Kenya's 47 counties between 2015 and 2025. Counties are labelled using Kenya's official county codes, with the corresponding county names provided in the accompanying key (Independent Electoral and Boundaries Commission, 2025).

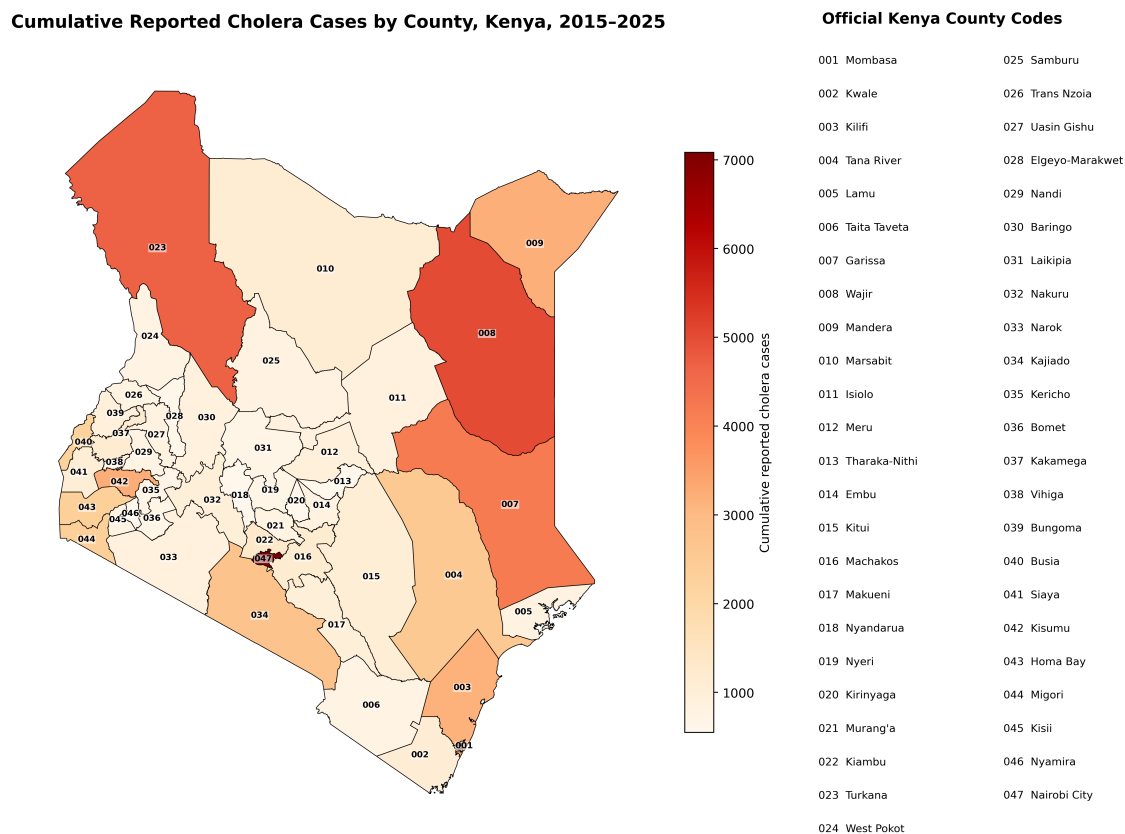


Figure 1.2: Spatial distribution of cumulative reported cholera cases across Kenya's 47 counties, 2015–2025. Counties are labelled using the official Kenya county codes, with corresponding county names shown in the accompanying key ([Independent Electoral and Boundaries Commission, 2025](#)). Cholera case totals were calculated from the study dataset. Administrative boundaries were sourced from the Independent Electoral and Boundaries Commission (IEBC) and distributed through OCHA's Humanitarian Data Exchange.

Spatial hotspot analyses are valuable for identifying locations with recurrent cholera burden and for guiding longer-term intervention prioritisation ([Global Task Force on Cholera Control, 2021](#); [Kiama et al., 2023](#); [Boru et al., 2023](#)). However, identifying historically vulnerable locations does not necessarily indicate when an outbreak is likely to occur ([Global Task Force on Cholera Control, 2021](#); [Kiama et al., 2023](#); [Dadi et al., 2025](#)). Short-term early warning addresses a different question: whether information available at the present time can identify an increased probability of outbreak occurrence during the following weeks ([Dadi et al., 2025](#); [Villanueva-Miranda et al., 2025](#)).

Combining recent epidemiological information with climatic, environmental and structural indicators may therefore provide a useful supplementary signal for preparedness ([Campbell et al., 2020](#); [Dadi et al., 2025](#)). Such a framework may be particularly useful where cholera risk varies both geographically and temporally, because effective early warning requires information on both where vulnerability is concentrated and when conditions associated with increased outbreak risk

are emerging (Global Task Force on Cholera Control, 2021; Kiama et al., 2023; Dadi et al., 2025; Villanueva-Miranda et al., 2025).

This study develops and evaluates a county-level early-warning framework for predicting cholera outbreaks two to four weeks in advance across Kenya's 47 counties. County-specific ARIMA models provide a statistical time-series baseline, while Random Forest and XGBoost are evaluated as machine-learning classifiers. Three weeks is used as the primary prediction horizon, with two- and four-week horizons examined as robustness analyses.

1.2 Problem Statement

Cholera outbreaks in Kenya continue to vary substantially across counties and over time despite extensive research describing historical burden, geographical hotspots, WASH vulnerability and outbreak-response challenges (Mutonga et al., 2013; Curran et al., 2018; Kiama et al., 2023; Boru et al., 2023). Much of this evidence is valuable for identifying vulnerable locations and informing longer-term planning, but comparatively less evidence is available on whether routinely available information can provide county-level warning several weeks before an outbreak (Dadi et al., 2025; Villanueva-Miranda et al., 2025).

Short-term prediction may benefit from combining recent disease activity with climatic, environmental and structural information (Campbell et al., 2020; Dadi et al., 2025). However, predictive models intended for early warning should be evaluated on future observations using temporally ordered training, validation and test periods to avoid optimistic performance estimates (Hyndman and Athanasopoulos, 2021; Dixon et al., 2022). Their usefulness should also be assessed in terms of both outbreaks detected and unnecessary alerts generated, rather than using a single performance measure alone (Hyndman and Athanasopoulos, 2021; Saito and Rehmsmeier, 2015; Dixon et al., 2022).

The study therefore addresses the limited evidence for a national, temporally evaluated and interpretable framework for multi-week county-level cholera outbreak prediction in Kenya.

1.3 Research Aim and Objectives

1.3.1 General Objective. To develop and evaluate a machine-learning early-warning framework for predicting county-level cholera outbreaks in Kenya using epidemiological, climatic, environmental, demographic, WASH and temporal indicators, while providing interpretable information on the predictors contributing to model predictions.

1.3.2 Specific Objectives.

1. To describe the temporal distribution of reported cholera cases, deaths, incidence and outbreak occurrence across Kenya's 47 counties from 2015 to 2025.

2. To compare county-level ARIMA, Random Forest and XGBoost models for predicting cholera outbreaks three weeks in advance.
3. To evaluate the ability of the predictive models to detect future outbreaks while limiting false alerts.
4. To identify and interpret the predictors contributing most strongly to the selected machine-learning model.
5. To assess the robustness of XGBoost predictive performance across two-, three- and four-week prediction horizons.

1.4 Research Questions

1. What temporal patterns characterise reported cholera cases, deaths, incidence and outbreak occurrence across Kenya's 47 counties from 2015 to 2025?
2. How do ARIMA, Random Forest and XGBoost compare in predicting county-level cholera outbreaks three weeks in advance?
3. How effectively do the predictive models detect future outbreaks while limiting false alerts?
4. Which predictors contribute most strongly to the selected machine-learning model?
5. How does XGBoost performance vary across two-, three- and four-week prediction horizons?

1.5 Significance of the Study

Hotspot analyses support longer-term geographical prioritisation but do not directly provide information on near-term outbreak risk ([Global Task Force on Cholera Control, 2021](#); [Kiama et al., 2023](#); [Boru et al., 2023](#)). This study complements such approaches by evaluating whether routinely available information can provide county-level warning several weeks in advance.

The study also compares a conventional time-series baseline with machine-learning approaches using chronological evaluation and explicitly considers the trade-off between outbreak detection and false-alert burden. The inclusion of model-interpretation methods further provides information on which variables contribute most strongly to the fitted predictions.

If prospectively validated, the resulting risk estimates could provide a supplementary signal for routine surveillance and epidemiological review ([Wiens et al., 2019](#)). They could help identify counties that may require closer assessment when predicted risk is elevated ([Wiens et al., 2019](#); [Ghassemi et al., 2021](#)). The model is not intended to replace routine surveillance, laboratory confirmation, outbreak investigation or public-health judgement ([Ghassemi et al., 2021](#); [Wiens et al., 2019](#)).

1.6 Scope of the Study

The study covers all 47 counties in Kenya using weekly observations from 2015–2025, with the county-week as the unit of analysis. The primary outcome is future outbreak status, defined analytically as at least five reported cholera cases in a county-week. This study-specific analytical threshold is not presented as an official surveillance definition; current cholera surveillance guidance also incorporates clinical and laboratory evidence ([Global Task Force on Cholera Control, 2024](#)). A sensitivity analysis additionally evaluates a broader definition of at least five reported cases or incidence of at least one case per 100,000 population.

Predictors cover epidemiological, climatic, environmental, demographic, WASH and temporal domains. ARIMA provides the statistical baseline, while Random Forest and XGBoost are evaluated as machine-learning classifiers. Three weeks is the primary prediction horizon, with two- and four-week horizons included for comparison. SHAP is used as a secondary interpretive method to examine the contribution of predictors to the selected machine-learning model.

The analysis focuses on prediction rather than causal inference. Mortality is summarised descriptively but is not modelled as a separate outcome. Individual-level prediction, mechanistic transmission modelling and prospective field deployment are outside the scope of the study.

1.7 Structure of the Thesis

Chapter 2 reviews the relevant literature on cholera epidemiology, environmental and structural risk factors, forecasting, machine learning and model interpretation. Chapter 3 describes the study data, outcome definition, predictor construction, modelling procedures and evaluation strategy. Chapter 4 presents and discusses the descriptive, predictive, robustness and model-interpretation results. Chapter 5 summarises the principal findings, limitations, recommendations and conclusions of the study.

2. Literature Review

2.1 Introduction

This chapter reviews literature relevant to short-term cholera outbreak prediction in Kenya. It covers cholera epidemiology, climatic and WASH-related risk, forecasting approaches, temporal validation, model evaluation and interpretability, before identifying the research gap addressed by this study.

2.2 Cholera Epidemiology and the Kenyan Context

Cholera remains concentrated in settings where access to safe water, sanitation and timely outbreak response is limited (World Health Organization, 2024; Ali et al., 2015; Lessler et al., 2018). Endemic countries are estimated to experience approximately 1.3–4.0 million cases and 21,000–143,000 deaths annually, with Sub-Saharan Africa contributing a substantial share of the global burden (Ali et al., 2015; World Health Organization, 2024; Lessler et al., 2018; Global Task Force on Cholera Control, 2017).

Kenya has experienced recurrent cholera outbreaks with substantial spatial and temporal variation. National surveillance documented outbreaks in urban, coastal, lake, arid and refugee settings (Mutonga et al., 2013). The 2014–2016 outbreak revealed constraints in laboratory capacity, staffing, supplies and coordination (George et al., 2016; Curran et al., 2018), while later studies identified persistent geographical differences in cholera burden and WASH vulnerability (Kiama et al., 2023; Boru et al., 2023). Investigation of the 2017 Nairobi outbreak also highlighted the influence of crowding, population mobility and mass gatherings (Ngere et al., 2024).

Hotspot analyses are useful for identifying locations with recurrent burden and guiding long-term intervention priorities (Global Task Force on Cholera Control, 2021; Kiama et al., 2023; Boru et al., 2023). Early-warning models address a different question by estimating whether outbreak risk is elevated in a future period (Dadi et al., 2025; Villanueva-Miranda et al., 2025). Short-term forecasting can therefore complement geographical prioritisation by adding information on near-term risk.

2.3 Climatic, Environmental and WASH-Related Risk

Cholera occurrence has been associated with climatic and environmental conditions, although these relationships vary across settings (Colwell, 1996; Reyburn et al., 2011; Jutla et al., 2013; Perez-Saez et al., 2017). Heavy rainfall and flooding may contaminate water sources or disrupt sanitation, while temperature may influence environmental conditions associated with *V. cholerae* (Cann et al., 2013; Jutla et al., 2013; Tompkins and Di Giuseppe, 2015).

The timing of climatic effects also varies between locations (Reyburn et al., 2011; Tompkins and Di Giuseppe, 2015; Campbell et al., 2020). Rainfall associations may depend on water access, sanitation, infrastructure and preceding environmental conditions, and different studies have reported different lag periods before cholera occurrence (Jutla et al., 2013; Perez-Saez et al., 2017; Campbell et al., 2020). These findings support testing lagged and rolling climatic predictors directly within the Kenyan setting rather than assuming that lag structures identified elsewhere will transfer unchanged (Reyburn et al., 2011; Campbell et al., 2020; Dixon et al., 2022).

WASH conditions reflect persistent structural vulnerability (Cowman et al., 2017; Lessler et al., 2018; Kiama et al., 2023), whereas recent reported cases capture current epidemiological activity (Kane et al., 2014; Dadi et al., 2025). Combining surveillance, climatic and structural information may therefore provide complementary signals for short-term prediction (Campbell et al., 2020; Dadi et al., 2025; Villanueva-Miranda et al., 2025).

Large-scale environmental products enable environmental conditions to be assessed consistently across broad geographic areas (Funk et al., 2015; Hersbach et al., 2020; Didan, 2015). CHIRPS provides precipitation estimates, ERA5 provides meteorological reanalysis data, and MODIS provides remotely sensed vegetation information (Funk et al., 2015; Hersbach et al., 2020; Didan, 2015). However, aggregation of these data to the county level may mask substantial spatial variation within counties; county-level values should therefore be interpreted as approximations of local environmental exposure (Boru et al., 2023).

2.4 Cholera Forecasting

Forecasting requires that predictors be available before the outcome being predicted. For an outcome at week $t + h$, predictor information should therefore come from week t or earlier to prevent leakage from future observations (Hyndman and Athanasopoulos, 2021; Dixon et al., 2022).

Previous cholera forecasting studies have used both statistical and machine-learning approaches. Climate-based prediction has been examined in Haiti (Tompkins and Di Giuseppe, 2015), machine-learning methods have been applied to climatic and environmental risk signals (Campbell et al., 2020), and seasonal weather relationships have been used for prediction in Tanzania (Leo, 2020). More recent work has combined surveillance and climatic information for early warning and explored explainable, multi-source infectious-disease prediction (Dadi et al., 2025; Jideani and Gerber, 2026; Jillahi et al., 2025; Inam, 2025; Villanueva-Miranda et al., 2025).

Performance from different studies is not directly comparable because spatial scale, outcome definition, predictor availability, forecast horizon and validation design differ substantially (Dixon et al., 2022; Inam, 2025; Villanueva-Miranda et al., 2025). Predictive performance should therefore be evaluated within the intended population and surveillance setting (Dixon et al., 2022; Wiens et al., 2019).

2.5 Statistical and Machine-Learning Approaches

Time-series models are useful forecasting benchmarks because disease counts observed over time may be serially dependent (Hyndman and Athanasopoulos, 2021; Kane et al., 2014). ARIMA represents this dependence through autoregressive, differencing and moving-average components and can therefore estimate how much future activity is predictable from historical case patterns (Hyndman and Athanasopoulos, 2021).

Machine-learning methods provide greater flexibility for modelling heterogeneous predictors, non-linear relationships and interactions (Breiman, 2001; Chen and Guestrin, 2016). This may be useful where climatic conditions, recent disease activity and structural vulnerability interact (Campbell et al., 2020; Dadi et al., 2025). However, flexible algorithms may also learn patterns that do not persist in later periods, making temporal evaluation essential (Dixon et al., 2022; Wiens et al., 2019).

Random Forest combines decision trees fitted to bootstrap samples and random subsets of predictors (Breiman, 2001). It can represent nonlinear relationships and interactions, while temporal information is incorporated through predictors such as lags and rolling summaries (Kane et al., 2014; Dixon et al., 2022).

XGBoost builds boosted decision trees sequentially and uses regularisation, shrinkage and sub-sampling to control model complexity (Chen and Guestrin, 2016). It also allows positive-class weighting, which may help address class imbalance, although weighting does not necessarily improve sensitivity at a fixed threshold (Chen and Guestrin, 2016; Saito and Rehmsmeier, 2015).

ARIMA, Random Forest and XGBoost therefore represent distinct modelling strategies: a statistical time-series baseline, a bagged-tree ensemble and a boosted-tree ensemble (Hyndman and Athanasopoulos, 2021; Breiman, 2001; Chen and Guestrin, 2016). Their comparison under the same temporal framework allows performance on future observations, rather than algorithmic complexity, to determine model usefulness (Kane et al., 2014; Dixon et al., 2022).

2.6 Temporal Validation and Prediction Horizon

Forecast evaluation should preserve the temporal order of observations (Hyndman and Athanasopoulos, 2021; Dixon et al., 2022). Random splitting of longitudinal data can introduce information from later periods into model development and may produce optimistic estimates of future performance (Kane et al., 2014; Dixon et al., 2022). Chronological partitioning more closely reflects prospective application by developing models on earlier periods and evaluating them on later observations (Hyndman and Athanasopoulos, 2021; Dixon et al., 2022).

A separate validation period permits selection of hyperparameters, class weights and classification thresholds without reference to the final test period, while an untouched later test period provides a stronger assessment of temporal generalisation (Dixon et al., 2022; Wiens et al., 2019).

The choice of prediction horizon involves a trade-off between predictive performance and the amount of operational lead time available for response (Hyndman and Athanasopoulos, 2021;

Wiens et al., 2019). Shorter horizons may retain stronger predictive information but allow less time for response, whereas longer horizons provide greater warning time but usually involve greater uncertainty (Tompkins and Di Giuseppe, 2015; Dadi et al., 2025; Villanueva-Miranda et al., 2025). Evaluating several nearby prediction horizons allows this trade-off to be assessed empirically (Hyndman and Athanasopoulos, 2021).

2.7 Evaluation of Outbreak Prediction

When outbreak observations are less frequent than non-outbreak observations, overall accuracy can conceal poor detection of the outbreak class (Saito and Rehmsmeier, 2015). Evaluation should therefore consider both successful outbreak detection and false-alert burden (Saito and Rehmsmeier, 2015; Wiens et al., 2019).

Sensitivity measures the proportion of outbreaks detected, specificity the proportion of non-outbreaks correctly classified, and precision the proportion of positive predictions that are true positives (Saito and Rehmsmeier, 2015). F1-score combines precision and sensitivity, while balanced accuracy gives equal weight to sensitivity and specificity (Saito and Rehmsmeier, 2015).

ROC AUC measures discrimination between positive and negative observations across thresholds, whereas PR AUC focuses on the precision–recall relationship and is particularly informative under class imbalance (Saito and Rehmsmeier, 2015). The Brier score measures the squared difference between predicted probabilities and observed binary outcomes (Brier, 1950).

Discrimination and calibration assess different aspects of probabilistic prediction (Saito and Rehmsmeier, 2015; Brier, 1950). A model may rank observations correctly while producing probabilities that do not correspond closely to observed event frequencies (Brier, 1950; Wiens et al., 2019). Calibration is therefore important when model probabilities are interpreted as absolute outbreak risks (Wiens et al., 2019; Stiglic et al., 2020).

Classification thresholds also affect operational performance (Wiens et al., 2019; Saito and Rehmsmeier, 2015). Lower thresholds generally increase outbreak detection but also increase false alerts, while higher thresholds tend to have the opposite effect (Saito and Rehmsmeier, 2015). Threshold selection should therefore consider the public- health consequences of both missed outbreaks and false alerts rather than relying only on overall accuracy or a conventional default threshold (Saito and Rehmsmeier, 2015; Wiens et al., 2019).

2.8 Explainable Machine Learning

Interpretability is important when predictive models support health-related decisions because performance metrics alone do not explain which information contributed to predictions (Stiglic et al., 2020; Wiens et al., 2019; Ghassemi et al., 2021). Examining predictor contributions can help determine whether fitted models rely on plausible or potentially problematic patterns (Lundberg and Lee, 2017; Stiglic et al., 2020).

SHapley Additive exPlanations (SHAP) apply Shapley values from cooperative game theory to allocate model-output contributions across predictors (Shapley, 1953; Lundberg and Lee, 2017). SHAP values can be summarised globally to describe overall predictor importance or examined locally for individual predictions (Lundberg and Lee, 2017). Explainable approaches have increasingly been used in infectious-disease modelling to improve transparency (Jideani and Gerber, 2026; Jillahi et al., 2025).

SHAP values describe the fitted prediction function rather than causal effects (Ghassemi et al., 2021; Wiens et al., 2019). A large SHAP contribution therefore indicates that a predictor influenced the model output but does not imply that changing that predictor would necessarily alter cholera occurrence (Ghassemi et al., 2021; Wiens et al., 2019).

2.9 Research Gap

Kenyan cholera research has documented historical outbreaks, geographical hotspots, WASH inequalities and outbreak-response constraints (Mutonga et al., 2013; Curran et al., 2018; Cowman et al., 2017; Kiama et al., 2023; Boru et al., 2023; Ngere et al., 2024). However, less evidence is available on national county-level prediction of cholera outbreaks several weeks in advance using a common temporally ordered framework (Kiama et al., 2023; Boru et al., 2023; Dadi et al., 2025).

Evidence from other settings suggests that epidemiological, climatic and environmental information can support short-term prediction (Tompkins and Di Giuseppe, 2015; Campbell et al., 2020; Leo, 2020; Dadi et al., 2025). However, the timing and strength of these associations differ between settings, and predictive performance cannot be assumed to generalise across populations or surveillance systems (Reyburn et al., 2011; Perez-Saez et al., 2017; Dixon et al., 2022; Wiens et al., 2019).

There is also limited Kenyan evidence comparing county-specific time-series baselines with tree-based machine-learning models under the same chronological development and independent future-test design (Kane et al., 2014; Dixon et al., 2022). This comparison is important because greater algorithmic complexity does not necessarily translate into better prospective performance (Dixon et al., 2022; Wiens et al., 2019).

The present study addresses these gaps by integrating epidemiological, climatic, environmental, demographic and WASH information across all 47 counties; comparing ARIMA, Random Forest and XGBoost under chronological evaluation; examining two-, three- and four-week prediction horizons; and using SHAP to interpret the selected machine-learning model.

2.10 Conclusion

The literature shows that short-term cholera risk may reflect recent epidemiological activity, climatic and environmental conditions, and persistent WASH-related vulnerability (Cowman et al.,

2017; Campbell et al., 2020; Dadi et al., 2025; Kiama et al., 2023). It also demonstrates the importance of chronological validation, appropriate evaluation under class imbalance, probabilistic assessment and transparent interpretation (Saito and Rehmsmeier, 2015; Brier, 1950; Lundberg and Lee, 2017; Dixon et al., 2022). These principles form the basis of the methodology described in Chapter 3.

3. Methodology

This chapter describes the study design, data, outcome and predictor construction, chronological partitioning, modelling, evaluation, robustness analyses and model interpretation.

3.1 Study Design and Analytical Framework

A longitudinal observational design was used to develop a county-level cholera early-warning framework for Kenya's 47 counties. Weekly observations from 2015–2025 formed a balanced panel of 574 observations per county, giving 26,978 county-week records.

The analysis addressed prediction rather than causal inference. For county i at epidemiological week t , let $\mathbf{X}_{i,t}$ denote the 23-dimensional predictor vector available at prediction time. For prediction horizon $h \in \{2, 3, 4\}$,

$$Y_{i,t}^{(h)} = Z_{i,t+h}, \quad p_{i,t}^{(h)} = P\left(Y_{i,t}^{(h)} = 1 \mid \mathbf{X}_{i,t}\right), \quad (3.1.1)$$

where $Z_{i,t+h}$ denotes future outbreak status. Three weeks was the primary horizon, while two and four weeks were evaluated to examine the trade-off between predictive performance and warning time (Tompkins and Di Giuseppe, 2015; Dadi et al., 2025; Villanueva-Miranda et al., 2025).

3.2 Data Sources and Harmonisation

The analytical dataset integrated county-week cholera surveillance, climatic, environmental, demographic and water, sanitation and hygiene (WASH) information for 2015–2025. Surveillance variables comprised reported cholera cases, deaths and incidence per 100,000 population. Routine surveillance data may be influenced by under-reporting, reporting delays, healthcare-seeking behaviour and variation in diagnostic confirmation (Mutonga et al., 2013; George et al., 2016; Curran et al., 2018).

The assembled dataset included rainfall, mean temperature, NDVI, flood and drought indicators, population and a county-level WASH vulnerability index. The environmental products used in constructing the study dataset were consistent with CHIRPS rainfall, ERA5 meteorological and MODIS vegetation products (Funk et al., 2015; Hersbach et al., 2020; Didan, 2015). Population was represented by the 2019 Kenya Population and Housing Census denominator (Kenya National Bureau of Statistics, 2019). Audit of the analytical dataset confirmed that each county's 2019 population value was held constant throughout 2015–2025 rather than interpolated annually.

The supplied study documentation did not provide sufficient upstream provenance to independently identify the original source of every surveillance, flood, drought and WASH variable. These variables are therefore reported as components of the assembled study dataset rather than being

assigned an unverified source. Kenyan studies nevertheless document marked geographical variation in WASH conditions and cholera burden (Cowman et al., 2017; Kiama et al., 2023; Boru et al., 2023).

All information was harmonised to the county-week unit before modelling. Climatic and environmental variables were represented at weekly county level, while population and WASH vulnerability were structural county-level variables.

Table 3.1: Principal data domains used in the analysis

Domain	Variables	Source/basis
Surveillance	Cases, deaths, incidence	Assembled county-week surveillance dataset
Climate	Rainfall, mean temperature	Environmental data products; CHIRPS and ERA5 documented in the study framework (Funk et al., 2015; Hersbach et al., 2020)
Environment	NDVI, flood, drought	MODIS vegetation product for NDVI (Didan, 2015); flood and drought variables from assembled study dataset
Population	2019 county population	Kenya Population and Housing Census (Kenya National Bureau of Statistics, 2019)
WASH	Vulnerability index (0–1)	Assembled study dataset

3.3 Outcome Definition

Weekly incidence was calculated as

$$I_{i,t} = \frac{C_{i,t}}{N_i} \times 100,000, \quad (3.3.1)$$

where $C_{i,t}$ is the reported number of cholera cases and N_i is the fixed 2019 population denominator for county i .

For the primary analysis,

$$Z_{i,t} = \mathbb{I}(C_{i,t} \geq 5). \quad (3.3.2)$$

Thus, a county-week containing at least five reported cases was classified as an analytical outbreak. The five-case threshold was adopted as a pragmatic study-specific criterion to distinguish

isolated low-count reporting from more substantial weekly cholera activity while retaining sufficient outbreak observations for predictive evaluation. This threshold is not an official surveillance definition of a suspected, probable or confirmed cholera outbreak, which may incorporate clinical, rapid-diagnostic and laboratory evidence ([Global Task Force on Cholera Control, 2024](#)). Sensitivity to a broader population-standardised rule was therefore assessed separately.

A sensitivity analysis evaluated a broader rule,

$$Z_{i,t}^* = \mathbb{I}(C_{i,t} \geq 5 \text{ or } I_{i,t} \geq 1 \text{ per } 100,000), \quad (3.3.3)$$

to determine whether conclusions were sensitive to incorporating a population-standardised incidence criterion.

3.4 Predictor Construction

The machine-learning matrix contained 23 predictors selected according to epidemiological relevance, availability at prediction time and support from previous cholera research ([Jutla et al., 2013](#); [Reyburn et al., 2011](#); [Campbell et al., 2020](#); [Dadi et al., 2025](#)). Future information was excluded to prevent target leakage.

Table 3.2: Predictor domains used in the machine-learning models

Domain	Predictors
Epidemiological	Current cases, incidence, current outbreak status
Temporal	Calendar year, epidemiological week, rainy-season indicator
Climate	Rainfall, temperature, rainfall lags 1–4, temperature lags 1–3, four- and eight-week rolling rainfall, rainfall–temperature interaction
Environmental	Flood indicator, drought index, NDVI
Structural	Population, WASH vulnerability index

Lagged variables were generated separately within each county after chronological ordering. Rainfall lags extended from one to four weeks and temperature lags from one to three weeks. Four- and eight-week rolling rainfall summaries and a rainfall–temperature interaction were also constructed. Future cases, incidence and outbreak indicators were excluded from the predictor matrix.

3.5 Missing Data and Analytical Sample

Missing values arising from lagged variables, rolling summaries and future targets occurred mainly at county-series boundaries. Complete-case filtering was applied only to records lacking a required

predictor or future target; no mean, median or model-based imputation was used for structurally undefined values.

For the primary three-week modelling analysis, 470 of 26,978 county-weeks were excluded, leaving 26,508 observations (98.26% of the original panel). No missing values remained in the final modelling matrix.

3.6 Chronological Partitioning

Model development preserved temporal order because random splitting of time-indexed health data may produce optimistic estimates of future performance (Hyndman and Athanasopoulos, 2021; Dixon et al., 2022). Partition assignment was based on the date of the future outcome rather than the predictor week.

Table 3.3: Chronological partition used for model development

Partition	Target period	Purpose
Training	2015–2022	Model fitting
Validation	2023–2024	Model and threshold selection
Test	2025	Independent evaluation

For horizon h , the target date was defined as

$$D_{i,t}^{(h)} = D_{i,t} + 7h \text{ days.} \quad (3.6.1)$$

This ensured that observations near calendar boundaries were allocated according to the period in which their outcomes occurred. The final three-week 2025 evaluation comprised 2,397 county-weeks, including 705 outbreak and 1,692 non-outbreak observations.

3.7 Statistical Time-Series Baseline

County-specific ARIMA models were used as the statistical baseline because they model temporal dependence in historical case series directly (Hyndman and Athanasopoulos, 2021; Kane et al., 2014). Models were fitted separately to each county's weekly cholera case series. Non-seasonal ARIMA models were compared with seasonal specifications using a 52-week seasonal period where appropriate.

Forecast case counts were converted to binary outbreak predictions using the same primary rule of at least five cases applied to the machine-learning outcomes. This ensured that ARIMA and machine-learning performance were compared using the same outcome definition and test observations.

3.8 Machine-Learning Models

Random Forest and XGBoost were selected as complementary tree-based classifiers capable of representing nonlinear associations and interactions (Breiman, 2001; Chen and Guestrin, 2016). Random Forest uses bagging across multiple decision trees, whereas XGBoost constructs boosted trees sequentially with regularisation to control complexity (Breiman, 2001; Chen and Guestrin, 2016).

Random Forest models with unweighted and balanced class settings were compared on the 2023–2024 validation period. XGBoost hyperparameters were selected sequentially on the same validation period while the 2025 test period remained untouched. Positive-class weights of 1, 2, 3, 4, 5, 6, and 8.07, maximum depths of 2, 3, 4, 5, and learning rates of 0.01, 0.03, 0.05, 0.07, and 0.10 were examined.

The final XGBoost specification was:

Table 3.4: Locked XGBoost specification

Parameter	Value	Parameter	Value
Estimators	300	Maximum depth	2
Learning rate	0.03	Subsample	0.80
Column sample	0.80	Positive-class weight	6
Objective	Binary logistic	Random seed	42

The locked specification was used for the final three-week model and for the two- and four-week horizon comparison.

3.9 Class Imbalance and Threshold Selection

Class imbalance was handled through model weighting rather than over-sampling or synthetic observations. The final XGBoost model used `scale_pos_weight=6`.

Predicted probabilities were converted to alerts using

$$\hat{Y}_{i,t}^{(h)} = \mathbb{I} \left(\hat{p}_{i,t}^{(h)} \geq c \right). \quad (3.9.1)$$

Thresholds from 0.10 to 0.60 in increments of 0.05 were evaluated on the validation data. Selection considered sensitivity, specificity, precision, F1-score, balanced accuracy, missed outbreaks and false alerts. A threshold of 0.50 was retained because it preserved moderate outbreak detection while substantially reducing false alerts compared with lower thresholds. The threshold was fixed before evaluation of the 2025 test period.

3.10 Performance and Calibration Assessment

Classification performance was evaluated using accuracy, sensitivity, specificity, precision, F1-score and balanced accuracy. ROC AUC assessed overall discrimination, while PR AUC was included because of class imbalance (Saito and Rehmsmeier, 2015). Probability error was assessed using the Brier score,

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2, \quad (3.10.1)$$

where lower values indicate smaller probabilistic prediction error (Brier, 1950).

The numbers of true alerts, false alerts and missed outbreaks were retained to support operational interpretation. Calibration of the final XGBoost probabilities was assessed using a calibration curve, expected calibration error and logistic recalibration intercept and slope.

Uncertainty in final test performance was assessed using 2,000 bootstrap resamples of the 2025 county-week predictions. Percentile 95% confidence intervals were obtained for accuracy, precision, sensitivity, specificity, F1-score, balanced accuracy, ROC AUC, PR AUC and Brier score.

3.11 Sensitivity and Robustness Analyses

Several analyses examined whether the main conclusion depended on modelling choices. First, the final XGBoost model was re-estimated without calendar year because calendar year showed a strong contribution in the initial SHAP analysis and could represent secular rather than transferable epidemiological information. Performance was compared using sensitivity, specificity, balanced accuracy, ROC AUC, PR AUC and Brier score.

Second, the broader outbreak definition incorporating either five reported cases or incidence of at least one case per 100,000 was evaluated on the same analytical framework.

Third, unweighted and balanced Random Forest models were examined to assess the unexpected reduction in sensitivity after class weighting. Predicted probability distributions among observed outbreak weeks were compared to determine whether weighting shifted scores relative to the 0.50 threshold.

Finally, XGBoost performance was evaluated at two-, three- and four-week horizons using the same locked model specification and an aligned set of 2,397 2025 county-week observations.

3.12 Paired Comparison with ARIMA

The final XGBoost and non-seasonal ARIMA predictions were compared on the same 2,397 2025 county-weeks using the same primary outbreak definition. McNemar's test was used to

assess whether the models differed in their paired classification errors. A paired bootstrap with 2,000 resamples estimated 95% confidence intervals for between-model differences in accuracy, sensitivity, specificity and balanced accuracy.

This comparison was interpreted in terms of differing operational error profiles rather than assuming that a statistically significant difference in classification errors implied overall superiority of one model.

3.13 Model Interpretation

SHapley Additive exPlanations (SHAP) were used to interpret the final XGBoost model (Shapley, 1953; Lundberg and Lee, 2017). For observation x_i ,

$$f(x_i) = \phi_0 + \sum_{j=1}^p \phi_j(x_i), \quad (3.13.1)$$

where ϕ_0 is the expected model output and $\phi_j(x_i)$ is the contribution assigned to predictor j . TreeExplainer was used to obtain global and observation-level contributions. Mean absolute SHAP values, beeswarm plots and dependence plots were used to examine predictor influence.

SHAP values describe the fitted model rather than causal effects; a strong contribution therefore indicates influence on the prediction and not that changing the predictor would necessarily change cholera occurrence (Ghassemi et al., 2021; Wiens et al., 2019).

3.14 Reproducibility, Assumptions and Limitations

All processing and modelling were implemented in Python using Google Colab. Stochastic procedures used a fixed random seed of 42. The final computational environment included Python 3.13.15, NumPy 2.1.3, pandas 2.2.3, scikit-learn 1.6.1, XGBoost 3.4.1, SHAP 0.52.0 and statsmodels 0.14.6. Additional package and system details are reported in Appendix A.4.

ARIMA assumes that temporal dependence can be represented through autoregressive, differencing and moving-average components (Hyndman and Athanasopoulos, 2021). Random Forest and XGBoost do not model temporal dependence automatically, so temporal information was supplied through current, lagged, rolling and calendar predictors (Breiman, 2001; Chen and Guestrin, 2016).

The study relied on aggregated routine surveillance data that may be affected by reporting delays and under-reporting (Mutonga et al., 2013; Curran et al., 2018). County-level environmental and WASH indicators may also conceal within-county heterogeneity. Model associations were therefore interpreted as predictive rather than causal.

3.15 Ethical Considerations

The study used aggregated county-level secondary data without individual patient identifiers. Analysis and reporting were conducted at county level. The model was developed as a research and decision-support framework rather than an automatic outbreak-declaration system. Operational application would require epidemiological verification, human oversight and prospective evaluation (Wiens et al., 2019; Ghassemi et al., 2021).

4. Results and Discussion

This chapter presents the descriptive epidemiology, model development, independent 2025 evaluation, robustness analyses, comparison with the ARIMA baseline and interpretation of the final XGBoost model. Results are organised around the study objectives and supported by the principal visual outputs from the analysis.

4.1 Descriptive Epidemiology and Temporal Distribution

The complete surveillance panel contained 26,978 county-week observations from all 47 Kenyan counties between 2015 and 2025, with 574 observations per county. Across the study period, 75,165 reported cholera cases and 1,081 reported deaths were recorded, corresponding to an overall case-fatality ratio of 1.44%. Using the primary analytical definition of at least five reported cases in a county-week, 4,514 county-weeks were classified as outbreak observations.

Cholera burden varied substantially over time. Reported cases increased from 5,859 in 2015 to 9,679 in 2017, declined to 2,802 in 2018, and increased again from 2023. The highest burden occurred in 2024, when 13,045 cases, 185 deaths, an incidence of 27.43 cases per 100,000 population and 880 outbreak county-weeks were recorded.

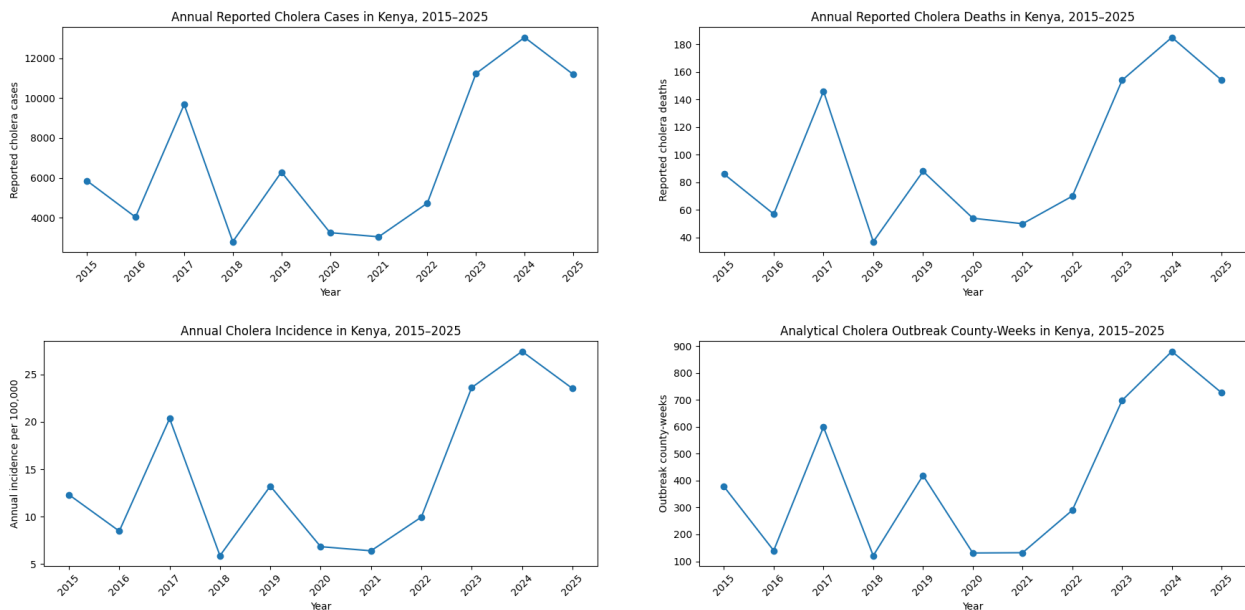


Figure 4.1: Annual temporal distribution of reported cholera cases, deaths, incidence and analytical outbreak county-weeks in Kenya, 2015–2025.

The annual patterns show that cases, deaths, incidence and analytical outbreak occurrence followed broadly similar temporal fluctuations. Outbreak county-weeks increased from 290 in 2022 to 698 in 2023 and peaked at 880 in 2024, representing 36.01% of county-week observations that

year. Although burden declined in 2025, 11,188 reported cases and 726 outbreak county-weeks were still observed. These findings answer the first study objective by demonstrating substantial year-to-year variation in cholera activity across the study period.

4.2 Development and Selection of the Machine-Learning Model

Construction of lagged predictors, rolling summaries and future outcomes reduced the primary three-week modelling sample from 26,978 to 26,508 county-weeks. Thus, 98.26% of the original panel was retained, with 23 predictors available for model development.

4.2.1 Random Forest Performance. Random Forest achieved high specificity but detected relatively few outbreak county-weeks during validation. Among 1,589 outbreak observations, the unweighted model detected 230 and missed 1,359, whereas the balanced model detected 111 and missed 1,478.

Table 4.1: Random Forest validation performance, 2023–2024

Metric	Unweighted	Balanced
Detected outbreaks	230	111
Missed outbreaks	1,359	1,478
False alerts	40	19
Precision	0.8519	0.8538
Sensitivity	0.1447	0.0699
Specificity	0.9879	0.9942
F1-score	0.2474	0.1291
Balanced accuracy	0.5663	0.5320
ROC AUC	0.7594	0.7582
PR AUC	0.6549	0.6454

The reduction in sensitivity after class weighting was examined using the distribution of predicted probabilities among observed outbreak weeks. The mean predicted probability decreased from 0.2805 in the unweighted model to 0.2275 in the balanced model, while the median declined from 0.2600 to 0.1933. Consequently, fewer outbreak observations exceeded the 0.50 classification threshold.

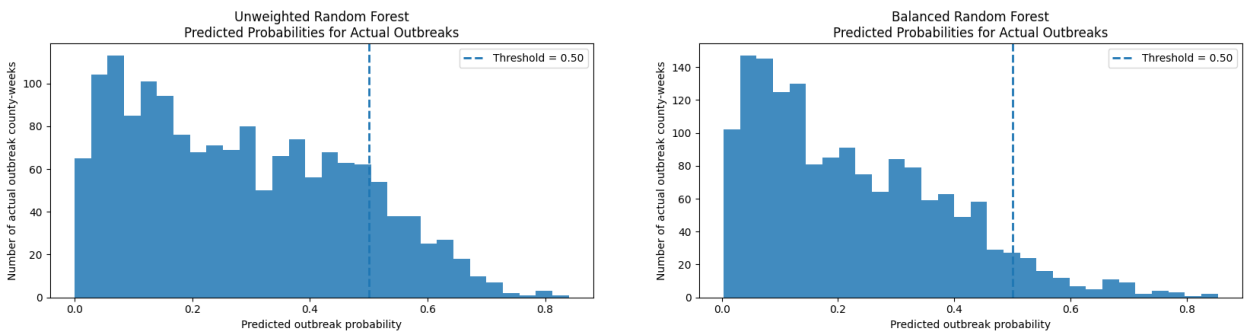


Figure 4.2: Predicted-probability distributions for the unweighted and balanced Random Forest models among observed outbreak county-weeks.

The diagnostic therefore showed that, in this fitted Random Forest, class weighting shifted outbreak probabilities downward relative to the fixed 0.50 threshold. Although specificity improved, the decline in outbreak detection made Random Forest less suitable for the intended early-warning purpose.

4.2.2 XGBoost Selection. XGBoost provided a more balanced relationship between outbreak detection and false alerts. The final validation specification used 300 estimators, maximum depth 2, learning rate 0.03, subsampling and column sampling of 0.80, and positive-class weight 6. At the 0.50 threshold, the selected model detected 1,014 of 1,589 outbreak observations, with sensitivity 0.6381, specificity 0.8312 and balanced accuracy 0.7346.

Table 4.2: Selected XGBoost validation performance

Measure	Value	Measure	Value
Actual outbreaks	1,589	Detected outbreaks	1,014
Missed outbreaks	575	False alerts	557
Precision	0.6454	Sensitivity	0.6381
Specificity	0.8312	F1-score	0.6418
Balanced accuracy	0.7346	PR AUC	0.6669

Detailed hyperparameter-search results are reported in Appendix [A.1](#).

4.3 Contribution of Predictor Domains

Predictor-group analysis compared climate/environment/WASH information, surveillance information and the combined model. Surveillance-only predictors detected the largest number of outbreaks but also generated the greatest false-alert burden. The combined model retained high outbreak detection while substantially reducing false alerts.

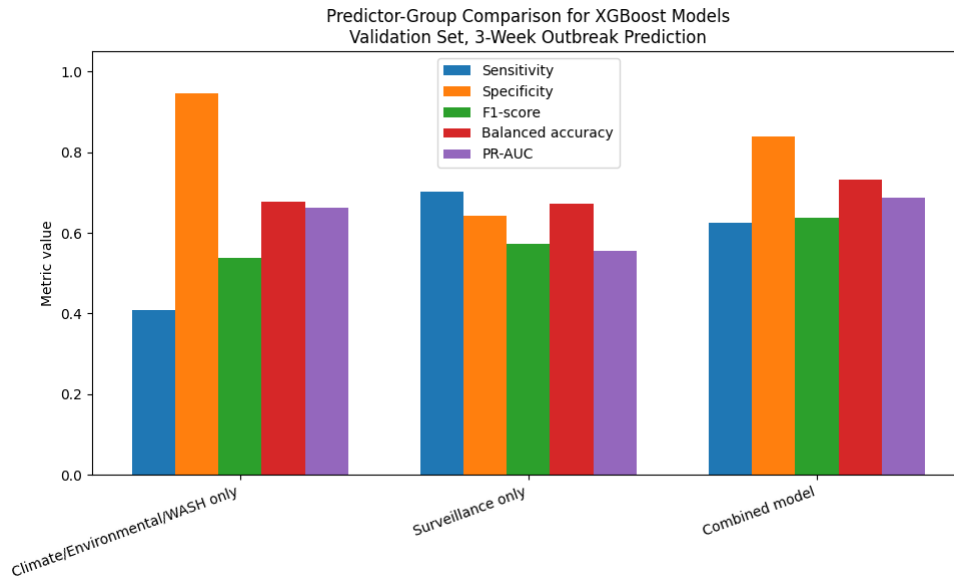


Figure 4.3: Validation performance of XGBoost models using different predictor domains for three-week outbreak prediction.

The climate/environment/WASH model detected 713 outbreaks and missed 876, whereas the surveillance-only model detected 1,077 but generated 1,112 false alerts. The combined model detected 1,014 outbreaks while reducing false alerts to 557. This suggests that recent surveillance and structural or environmental information contributed complementary predictive information (Jutla et al., 2013; Campbell et al., 2020; Dadi et al., 2025). The contribution of WASH-related information is also consistent with geographical inequalities in cholera vulnerability previously reported in Kenya (Cowman et al., 2017; Kiama et al., 2023; Boru et al., 2023).

4.4 Threshold Selection

The probability threshold determined the trade-off between detecting outbreak county-weeks and issuing unnecessary alerts.

Table 4.3: Selected XGBoost validation operating points

Threshold	Sensitivity	Specificity	False alerts
0.30	0.8213	0.5056	1,631
0.40	0.7338	0.6878	1,030
0.50	0.6381	0.8312	557
0.60	0.5582	0.8881	369

At a threshold of 0.30, sensitivity reached 0.8213 but 1,631 false alerts were generated. Increasing the threshold to 0.50 reduced false alerts to 557 while retaining sensitivity of 0.6381. At 0.60, false

alerts declined further to 369, but sensitivity fell to 0.5582. The threshold of 0.50 was therefore selected on validation data as an operational compromise rather than simply as a conventional default.

4.5 Temporal Stability

Expanding-window evaluation across 2017–2024 showed that model performance varied over time. Mean sensitivity was 0.5202, specificity 0.8062, balanced accuracy 0.6632, ROC AUC 0.7413 and PR AUC 0.3937.

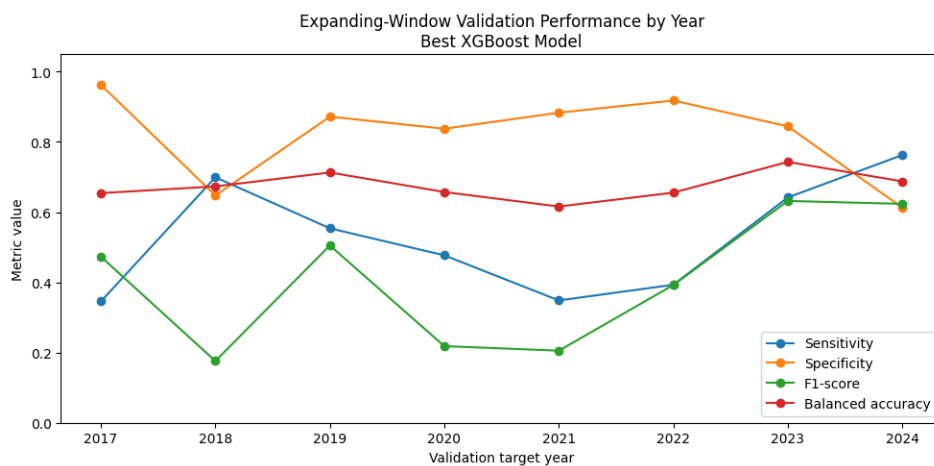


Figure 4.4: Year-specific XGBoost performance under expanding-window evaluation, 2017–2024.

The variation across years indicates that predictive performance was not temporally constant. This supports the need for prospective reassessment and periodic model updating rather than assuming that historical performance will remain unchanged (Dixon et al., 2022; Villanueva-Miranda et al., 2025).

4.6 Independent 2025 XGBoost Performance

Following model and threshold selection, the final XGBoost model was evaluated on the independent 2025 test period. The test set comprised 2,397 county-weeks, including 705 outbreak and 1,692 non-outbreak observations. The model detected 413 outbreaks and missed 292, while correctly classifying 1,441 non-outbreak county-weeks and generating 251 false alerts.

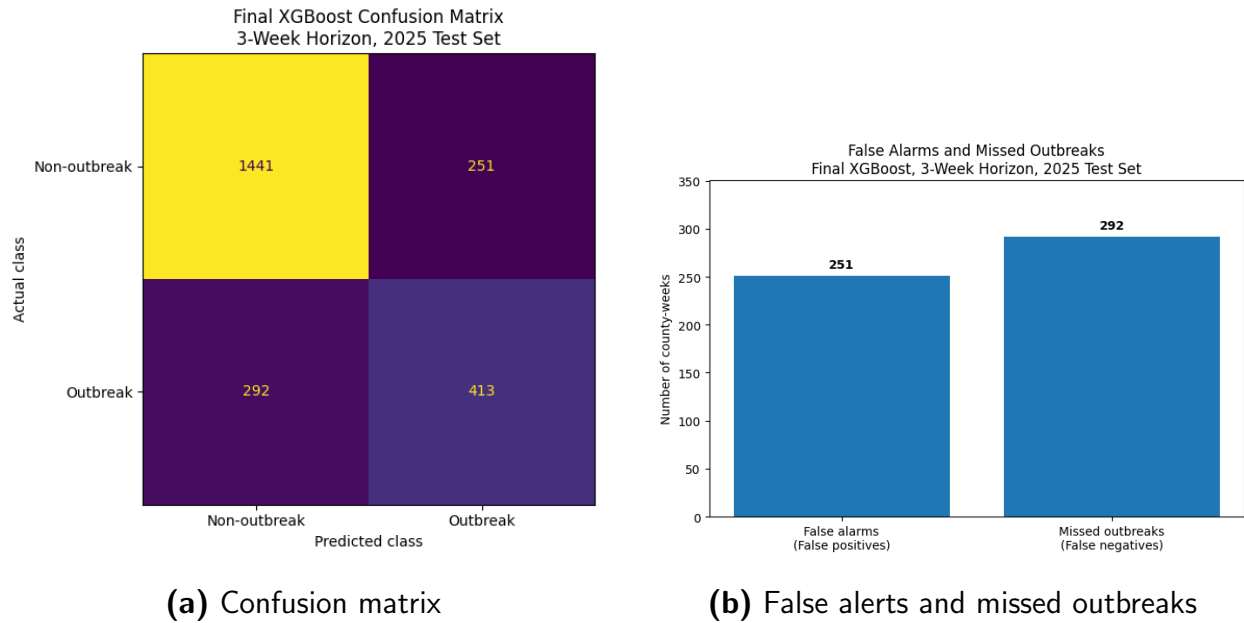


Figure 4.5: Classification performance of the final three-week XGBoost model on the independent 2025 test period. Panel (a) shows the confusion matrix, while panel (b) compares the numbers of missed outbreaks and false alerts.

Table 4.4: Independent 2025 performance of the final XGBoost model

Metric	Estimate	95% bootstrap CI
Accuracy	0.7735	0.7560–0.7902
Precision	0.6220	0.5846–0.6597
Sensitivity	0.5858	0.5497–0.6211
Specificity	0.8517	0.8341–0.8693
F1-score	0.6034	0.5716–0.6336
Balanced accuracy	0.7187	0.6987–0.7382
ROC AUC	0.7601	0.7363–0.7829
PR AUC	0.5933	0.5527–0.6303
Brier score	0.1741	0.1666–0.1817

Sensitivity of 0.5858 indicates that the model detected 58.6% of outbreak county-weeks, while 41.4% were missed. Precision of 0.6220 indicates that 62.2% of generated alerts corresponded to observed outbreak county-weeks. The 251 false alerts represented 14.8% of all non-outbreak county-weeks and 37.8% of all alerts generated during the test period.

Across the 47 counties, the model generated an average of 14.13 alerts and 5.34 false alerts per county during 2025. Overall, these results indicate moderate predictive ability rather than complete outbreak detection and illustrate the operational trade-off between detecting additional outbreaks and generating unnecessary alerts.

4.7 Probability Calibration

The mean predicted outbreak probability was 0.3989, compared with an observed outbreak prevalence of 0.2941. The expected calibration error was 0.1047, while logistic recalibration produced an intercept of -0.5702 and slope of 1.6499.

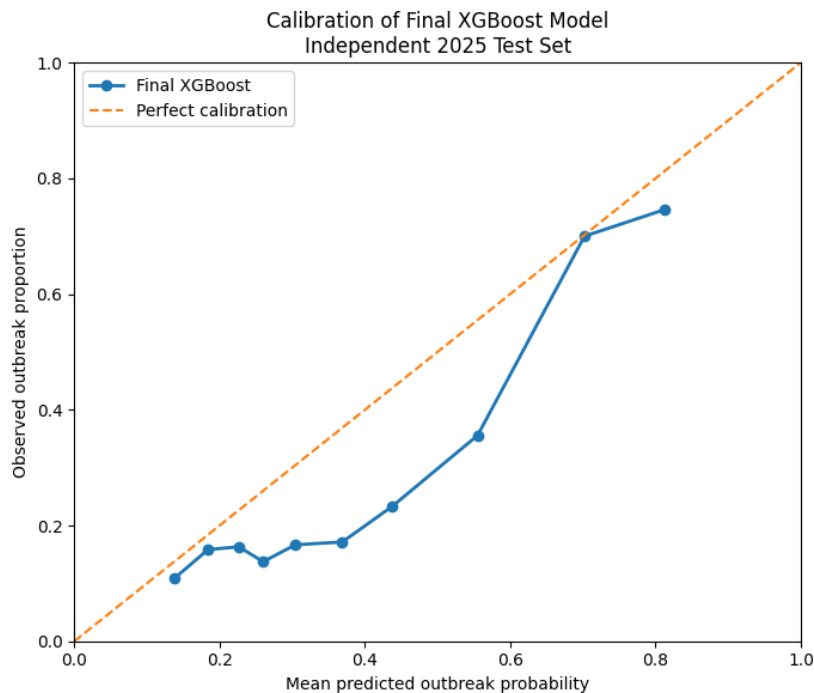


Figure 4.6: Calibration of predicted outbreak probabilities for the final three-week XGBoost model on the independent 2025 test period.

The calibration curve showed that predicted probabilities were generally higher than observed event frequencies across several intermediate-risk groups, while agreement improved at higher predicted risks. The negative intercept is consistent with overall overprediction, and the slope above one indicates that fitted probabilities were relatively compressed towards intermediate values. Thus, the model showed useful discrimination but its raw probabilities should not be interpreted as perfectly calibrated estimates of absolute outbreak risk.

4.8 Sensitivity to the Outbreak Definition

The broader analytical rule of at least five reported cases or incidence of at least one case per 100,000 increased the number of 2025 outbreak county-weeks from 705 to 730.

Sensitivity remained almost unchanged, whereas specificity and balanced accuracy declined modestly and the Brier score increased. The principal predictive conclusion was therefore reasonably robust to the broader analytical definition, although the alternative rule produced a larger false-alert burden.

Table 4.5: Sensitivity of XGBoost performance to the outbreak definition

Metric	Primary rule	Broader rule
Outbreak county-weeks	705	730
Sensitivity	0.5858	0.5863
Specificity	0.8517	0.8266
Balanced accuracy	0.7187	0.7065
ROC AUC	0.7601	0.7512
PR AUC	0.5933	0.5901
Brier score	0.1741	0.1812

4.9 Sensitivity to Calendar Year

Calendar year had the largest mean absolute SHAP contribution in the primary model, raising concern that the model might be learning secular temporal differences rather than only transferable epidemiological relationships. The final XGBoost model was therefore refitted without calendar year.

Removing calendar year increased sensitivity from 0.5858 to 0.7433, balanced accuracy from 0.7187 to 0.7283, ROC AUC from 0.7601 to 0.8148 and PR AUC from 0.5933 to 0.7024. However, specificity declined from 0.8517 to 0.7134, false alerts increased from 251 to 485 and the Brier score increased from 0.1741 to 0.1892.

The model therefore did not depend on calendar year alone. Removing this variable produced a more sensitive but substantially less specific warning profile, indicating that temporal transportability should be monitored carefully during prospective use.

4.10 Prediction-Horizon Robustness

The locked XGBoost specification was evaluated at two-, three- and four-week horizons using the same 2,397 test observations.

Table 4.6: XGBoost performance across prediction horizons in 2025

Horizon	Acc.	Sens.	Spec.	Bal. Acc.	ROC	PR
2 weeks	0.7722	0.5915	0.8475	0.7195	0.7601	0.5816
3 weeks	0.7735	0.5858	0.8517	0.7187	0.7601	0.5933
4 weeks	0.7543	0.5844	0.8251	0.7047	0.7317	0.5700

Performance at two and three weeks was nearly identical, with balanced accuracy differing by only 0.0008. The three-week horizon therefore provided one additional week of warning with negligible loss in balanced accuracy. Performance deteriorated more clearly at four weeks, where balanced accuracy fell to 0.7047 and ROC AUC to 0.7317. This pattern illustrates the expected trade-off

between increasing warning time and predictive uncertainty (Tompkins and Di Giuseppe, 2015; Dadi et al., 2025; Villanueva-Miranda et al., 2025).

4.11 Comparison with the ARIMA Baseline

The final non-seasonal ARIMA and XGBoost models were compared on the same 2,397 observations from the 2025 test period using the same primary outbreak definition.

Table 4.7: ARIMA and XGBoost performance on the 2025 test period

Metric	ARIMA	XGBoost
True positives	363	413
False negatives	342	292
True negatives	1,540	1,441
False positives	152	251
Accuracy	0.7939	0.7735
Precision	0.7049	0.6220
Sensitivity	0.5149	0.5858
Specificity	0.9102	0.8517
F1-score	0.5951	0.6034
Balanced accuracy	0.7125	0.7187

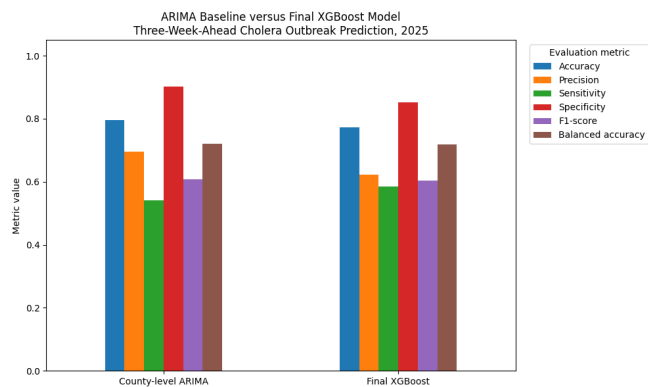


Figure 4.7: ARIMA versus XGBoost performance on the 2025 three-week target.

XGBoost detected 50 more outbreak county-weeks than ARIMA but generated 99 additional false alerts. ARIMA achieved higher accuracy, precision and specificity, whereas XGBoost achieved greater sensitivity and slightly higher balanced accuracy.

Paired bootstrap analysis estimated the XGBoost–ARIMA difference in sensitivity as 0.0708 (95% CI: 0.0453–0.0966) and the difference in specificity as -0.0586 (95% CI: -0.0731 to -0.0450). The corresponding difference in balanced accuracy was 0.0061 (95% CI: -0.0089 to 0.0206), with the confidence interval including zero. McNemar's test indicated that the two models differed in their paired classification errors ($p = 0.0017$).

Overall, the comparison indicates different operational error profiles rather than clear overall superiority of either approach. ARIMA was more conservative and generated fewer false alerts, whereas XGBoost detected more future outbreaks and incorporated information from multiple predictor domains.

4.12 Interpretation of the Final XGBoost Model

SHAP analysis was used as a secondary interpretive tool to examine which predictors contributed most strongly to the final XGBoost predictions (Lundberg and Lee, 2017). The analysis describes

how predictor values influenced the fitted model output and should therefore be interpreted as predictive attribution rather than evidence of causal effects (Ghassemi et al., 2021; Wiens et al., 2019).

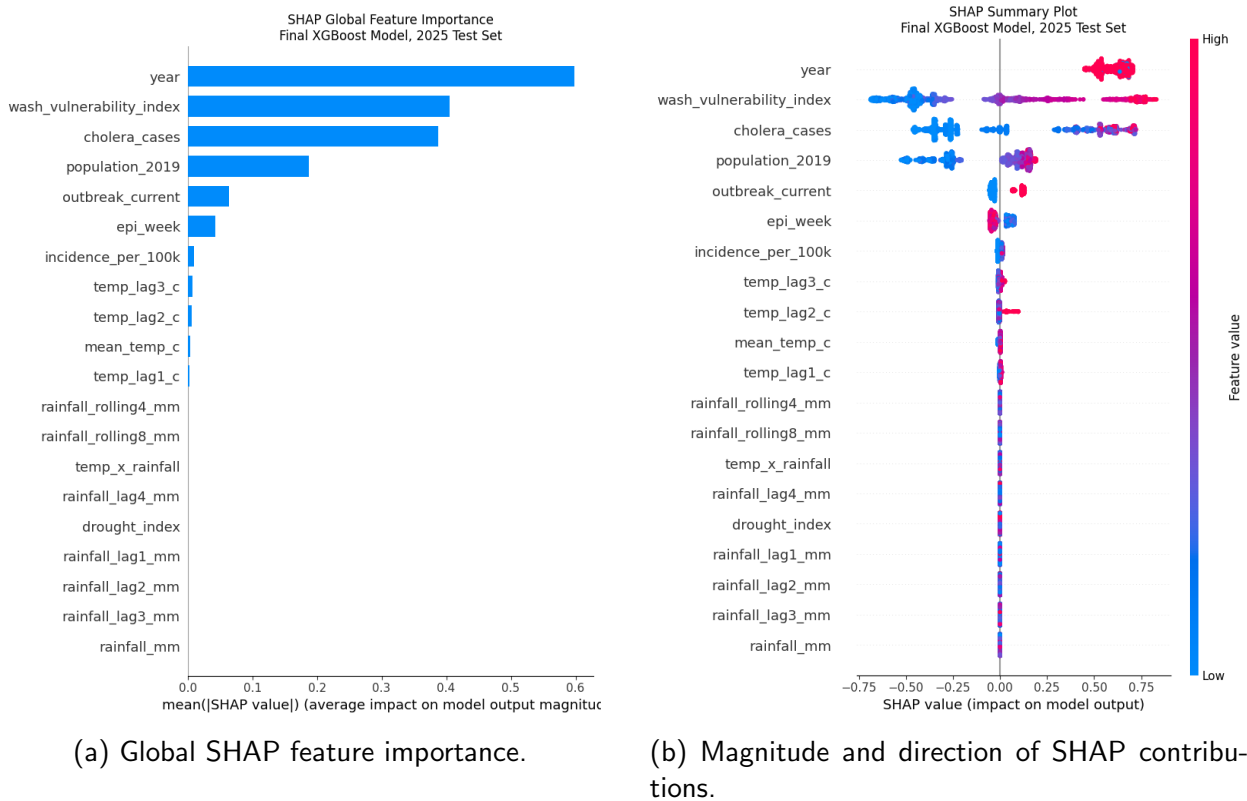


Figure 4.8: SHAP interpretation of the final three-week XGBoost model on the 2025 test period. Panel (a) ranks predictors according to mean absolute SHAP value, while panel (b) shows the direction and distribution of individual feature contributions.

Calendar year had the largest mean absolute SHAP value, followed by WASH vulnerability, current cholera cases and population. The prominence of calendar year motivated the no-year sensitivity analysis reported earlier and indicated the need to determine whether model discrimination depended primarily on long-term temporal information.

Recent cholera activity was also an important predictive signal. This is consistent with the role of recent surveillance information in short-term infectious-disease forecasting (Kane et al., 2014; Dadi et al., 2025). The contribution of WASH vulnerability is also consistent with Kenyan evidence documenting geographical inequalities in water and sanitation conditions and their relationship with cholera burden (Cowman et al., 2017; Kiama et al., 2023; Boru et al., 2023).

In the SHAP distribution plot, positive SHAP values increased the fitted outbreak score, whereas negative values reduced it. The variation in SHAP contributions across observations indicates that the influence of individual predictors was not constant throughout the dataset and is consistent with the nonlinear structure of the fitted XGBoost model (Chen and Guestrin, 2016; Lundberg

and Lee, 2017). These contributions describe the behaviour of the prediction model and should not be interpreted as causal effects (Ghassemi et al., 2021; Wiens et al., 2019).

WASH vulnerability and current cholera activity. The SHAP dependence plots provide additional detail on two of the most influential non-temporal predictors.

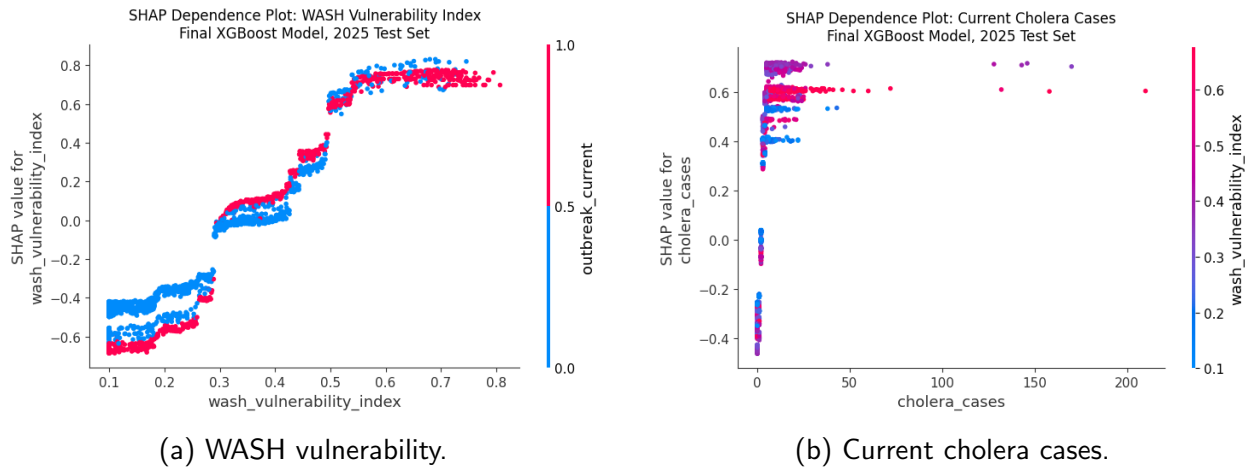


Figure 4.9: SHAP dependence relationships for selected predictors in the final three-week XGBoost model. Panel (a) shows WASH vulnerability and panel (b) shows current reported cholera cases.

Higher WASH vulnerability generally corresponded to more positive SHAP contributions, although the fitted relationship was nonlinear and varied across observations. This pattern is consistent with evidence that inadequate water and sanitation conditions are associated with greater cholera vulnerability in Kenya (World Health Organization, 2024; Cowman et al., 2017; Kiama et al., 2023; Boru et al., 2023). The SHAP relationship, however, reflects model attribution and should not be interpreted as evidence that the WASH variable independently caused future outbreaks.

Higher current cholera case counts also generally corresponded to more positive SHAP contributions and therefore higher fitted outbreak scores three weeks later. This finding is consistent with the importance of recent disease activity in infectious-disease forecasting (Kane et al., 2014; Dadi et al., 2025). The use of current cases does not constitute target leakage because the predictor represented information available at prediction week t , whereas the outcome represented outbreak status at $t + 3$.

4.13 Overall Interpretation

The results demonstrate a consistent trade-off between outbreak detection and false-alert burden. Random Forest achieved very high specificity but detected few outbreaks, whereas XGBoost produced a more useful balance for early warning. On the independent 2025 test period, XGBoost detected 58.6% of outbreak county-weeks while correctly classifying 85.2% of non-outbreak county-weeks.

The predictor-domain analysis showed that surveillance information was highly informative but produced many false alerts when used alone. Combining epidemiological, climatic, environmental and structural information reduced this burden while preserving substantial outbreak detection. Performance also varied across time, indicating that prospective monitoring and periodic model updating would be necessary.

The prediction-horizon analysis supported three weeks as the primary warning horizon. Performance at two weeks was almost identical, while four-week discrimination deteriorated more clearly. The three-week horizon therefore offered additional preparation time without a meaningful reduction in balanced accuracy compared with two weeks.

The comparison with ARIMA further showed that machine learning was not uniformly superior. ARIMA produced fewer false alerts and greater specificity, whereas XGBoost detected more outbreaks. The principal contribution of XGBoost was therefore its more sensitive warning profile and its ability to integrate multiple predictor domains rather than dominance across every performance measure.

Overall, the findings demonstrate moderate predictive ability for county-level cholera outbreak risk three weeks in advance. The framework is best interpreted as a supplementary early-warning signal that could prompt closer review of surveillance reports and heightened epidemiological assessment. Laboratory confirmation, routine surveillance and public-health judgement would remain essential before operational response decisions are made.

5. Conclusion

This chapter summarises the principal findings, scientific contribution, limitations and practical implications of the study, and outlines priorities for future research.

5.1 Summary of Key Findings

This study developed and evaluated a county-level early-warning framework for short-term cholera outbreak prediction in Kenya using weekly epidemiological, climatic, environmental, demographic and WASH-related information from 2015–2025.

The descriptive analysis showed substantial temporal variation in cholera burden across the study period. A total of 75,165 reported cases and 1,081 deaths were recorded, while 4,514 county-weeks met the primary analytical outbreak definition. The greatest annual burden occurred in 2024, with 13,045 cases, 185 deaths and 880 outbreak county-weeks.

Among the machine-learning approaches, Random Forest achieved high specificity but detected relatively few outbreaks. XGBoost provided a more balanced warning profile. The final three-week model detected 413 of 705 outbreak county-weeks in the independent 2025 test period and generated 251 false alerts among 1,692 non-outbreak observations. Sensitivity was 0.5858, specificity 0.8517, balanced accuracy 0.7187, ROC AUC 0.7601 and PR AUC 0.5933.

Performance at two and three weeks was nearly identical, whereas a clearer decline occurred at four weeks. The three-week horizon was therefore retained because it provided one additional week of warning compared with two weeks with negligible loss in balanced accuracy.

The ARIMA comparison demonstrated that machine learning was not uniformly superior. ARIMA produced fewer false alerts and higher specificity, while XGBoost detected 50 additional outbreak county-weeks. The paired comparison therefore indicated different operational error profiles rather than clear overall superiority of either approach.

The final XGBoost probabilities also showed imperfect calibration. Predicted risk tended to exceed observed outbreak frequency in several intermediate-risk groups, indicating that the model probabilities should not be interpreted as fully calibrated estimates of absolute risk.

SHAP analysis showed that temporal information, recent cholera activity and structural vulnerability contributed strongly to model predictions. Calendar year had the largest mean absolute SHAP contribution. However, sensitivity analysis excluding calendar year showed that predictive discrimination did not depend on this variable alone, although the resulting model generated substantially more false alerts.

The broader outbreak-definition sensitivity analysis also produced similar sensitivity and only modest changes in discrimination, suggesting that the main predictive conclusions were reasonably robust to the alternative analytical rule.

The principal contribution of this study is therefore the development of a temporally evaluated

and interpretable county-level forecasting framework that integrates several predictor domains, compares statistical and machine-learning approaches on the same future test observations, and quantifies the practical trade-off between outbreak detection and false alerts.

5.2 Study Limitations

Several limitations should be considered when interpreting the findings. First, the analysis relied on routine aggregated surveillance data, which may be affected by under-reporting, reporting delays, healthcare-seeking behaviour and variation in diagnostic confirmation.

Second, environmental, population and WASH information was represented at county level. These summaries may conceal important within-county differences in exposure, transmission and access to water and sanitation. The fixed 2019 population denominator used throughout 2015–2025 may also differ from actual county populations in individual years.

Third, human mobility, cross-border transmission, laboratory-confirmation patterns, intervention coverage and local changes in water and sanitation infrastructure were not modelled directly.

Fourth, the final independent test period covered only 2025. Expanding-window evaluation and the calendar-year sensitivity analysis both showed that model performance can vary over time. Generalisation to future years should therefore not be assumed without prospective evaluation.

Finally, the bootstrap confidence intervals were based on resampling county-week observations and did not explicitly model within-county temporal dependence. The resulting uncertainty estimates should therefore be interpreted as approximate measures of sampling variability.

5.3 Future Research

Future work should prioritise prospective evaluation using newly arriving surveillance data to determine whether the observed performance is maintained under real-time conditions. Particular attention should be given to temporal stability, calibration drift and changes in the balance between outbreak detection and false-alert burden.

Because the present analysis identified imperfect probability calibration, future studies should examine recalibration methods and assess whether calibration remains stable as new data accumulate. Ensemble approaches that combine the complementary error profiles of ARIMA and XGBoost could also be investigated.

Where reliable data are available, future analyses could extend prediction to finer spatial scales such as sub-counties or health-facility catchment areas. Additional predictors including human mobility, laboratory confirmation, water quality, intervention coverage, socioeconomic conditions and cross-border outbreak activity may further improve forecasting if they are available consistently before the prediction target.

Future studies should also evaluate uncertainty using resampling or modelling approaches that

explicitly account for temporal and within-county dependence.

5.4 Practical and Policy Recommendations

The framework should undergo prospective validation before routine operational use. If performance remains acceptable, county-week risk predictions could be reviewed alongside routine surveillance to identify locations requiring closer examination of recent case reports and enhanced epidemiological assessment.

Predicted alerts should not be treated as automatic outbreak declarations. Routine surveillance, laboratory confirmation and epidemiological judgement should remain central to public-health decision-making. The model is therefore best considered an additional decision-support signal rather than a replacement for existing surveillance systems.

Because performance varied across time, future implementation should include regular performance monitoring, recalibration where necessary and periodic model updating. Classification thresholds should also be reviewed in relation to available response capacity and the relative consequences of missed outbreaks and unnecessary alerts.

5.5 Concluding Remarks

The findings demonstrate moderate predictive ability for county-level cholera outbreak risk three weeks in advance using routinely available epidemiological, climatic, environmental and structural information.

XGBoost provided a more sensitive warning profile, while ARIMA generated fewer false alerts and achieved greater specificity. The choice between these approaches therefore depends on the operational importance assigned to missed outbreaks and false alerts rather than on a single performance measure.

With prospective validation, periodic updating, calibration monitoring and continued epidemiological oversight, the proposed framework may provide a useful supplementary signal for earlier recognition of elevated cholera risk and support preparedness and surveillance decision-making in Kenya.

Appendix A. Supplementary Material

This appendix provides supplementary model-development and reproducibility information supporting the analyses reported in Chapter 4. It includes detailed XGBoost tuning results, additional threshold information, temporal robustness summaries and the computational environment used for the final analysis.

A.1 XGBoost Hyperparameter Evaluation

XGBoost tuning was conducted using the 2023–2024 validation period, while the 2025 test period remained unused. Positive-class weighting, tree depth and learning rate were evaluated sequentially before the final model specification was locked.

A.1.1 Tree Depth. Maximum depths of 2, 3, 4 and 5 were compared with positive-class weight fixed at 6.

Table A.1: Validation performance across XGBoost tree depths

Depth	Detected	Missed	False	Sens.	Spec.	ROC	PR
2	991	598	567	0.6237	0.8281	0.7736	0.6579
3	924	665	547	0.5815	0.8342	0.7513	0.6146
4	843	746	476	0.5305	0.8557	0.7450	0.5981
5	743	846	429	0.4676	0.8700	0.7337	0.5833

Depth 2 achieved the highest sensitivity, ROC AUC and PR AUC among the tested values and was therefore retained.

A.1.2 Learning Rate. With maximum depth fixed at 2 and positive-class weight at 6, learning rates from 0.01 to 0.10 were evaluated.

Table A.2: Validation performance across XGBoost learning rates

Rate	Detected	False	Sens.	Spec.	F1	ROC	Brier
0.01	996	546	0.6268	0.8345	0.6362	0.7874	0.1746
0.03	1,014	557	0.6381	0.8312	0.6418	0.7776	0.1760
0.05	991	567	0.6237	0.8281	0.6298	0.7736	0.1750
0.07	985	579	0.6199	0.8245	0.6248	0.7644	0.1776
0.10	959	610	0.6035	0.8151	0.6073	0.7567	0.1806

Although a learning rate of 0.01 achieved the highest ROC AUC and lowest Brier score, 0.03 produced the highest sensitivity, F1-score and number of detected outbreaks. It was therefore selected for the early-warning objective.

A.1.3 Positive-Class Weight. Candidate positive-class weights were

1, 2, 3, 4, 5, 6, 8.07,

where 8.07 approximated the negative-to-positive ratio in the development data. A weight of 6 was retained because it provided a more useful balance between outbreak detection and false alerts than the larger weight. Under an earlier depth-four specification, weight 8.07 generated 685 false alerts with precision 0.5764, compared with 476 false alerts and precision 0.6391 at weight 6.

A.2 Probability-Threshold Evaluation

Thresholds from 0.10 to 0.60 were evaluated using validation data before the 2025 test period was examined. Selected operating points are shown below.

Table A.3: Selected XGBoost validation thresholds

Threshold	Sensitivity	Specificity	False alerts
0.30	0.8213	0.5056	1,631
0.40	0.7338	0.6878	1,030
0.50	0.6381	0.8312	557
0.60	0.5582	0.8881	369

The 0.50 threshold was retained because it substantially reduced false alerts relative to lower thresholds while preserving moderate outbreak detection. It was fixed before independent 2025 evaluation.

A.3 Expanding-Window Evaluation

Temporal robustness was assessed using eight expanding-window annual evaluation folds between 2017 and 2024.

Performance varied between evaluation years, supporting continued monitoring and periodic model updating during prospective application.

A.4 Computational Reproducibility

All data processing, feature construction, modelling and statistical evaluation were implemented in Python using Google Colab. The computational environment and software versions recorded during the final analytical workflow are summarised in Table [A.5](#).

Table A.4: Mean expanding-window performance, 2017–2024

Metric	Mean
Sensitivity	0.5202
Specificity	0.8062
F1-score	0.4001
Balanced accuracy	0.6632
ROC AUC	0.7413
PR AUC	0.3937
Brier score	0.1635

A fixed random seed of 42 was used for stochastic model-fitting procedures and bootstrap re-sampling where applicable. Predictor definitions, chronological data partitions and the locked XGBoost specification were retained throughout the final evaluation.

Table A.5: Computational environment recorded for the final analysis

Component	Version or environment
Python	3.13.15
Operating system	Linux 6.6.122+ (x86_64)
Computing environment	Google Colab
NumPy	2.1.3
pandas	2.2.3
scikit-learn	1.6.1
XGBoost	3.4.1
SHAP	0.52.0
statsmodels	0.14.6
pmdarima	2.1.1
Matplotlib	3.10.0
SciPy	1.16.3
Random seed	42

The final XGBoost model used 300 estimators, maximum tree depth 2, learning rate 0.03, sub-sample fraction 0.80, column-subsample fraction 0.80 and positive-class weight 6. A classification threshold of 0.50 was fixed before independent evaluation, and the primary prediction horizon was three weeks. Bootstrap uncertainty estimates were based on 2,000 resamples.

Together with the fixed random seed, chronological partitioning and locked model specification, these details provide the principal information required to reproduce the final modelling workflow under a comparable computational environment.

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